SYNTAX ERROR 2026
OCT23-OCT25

PROBLEM STATEMENT - ALUM TRACK DEV I

Do You Know How to Train?
Train an agent that can catch a train.

The Challenge:
Build one general-purpose agent that uses a computer the way a person does: terminal, files, browser and desktop apps. Then teach it specific jobs through **skills**, not new code.
The final exam is the most stressful computer task most Indians know: booking a Tatkal ticket the second the window opens.

The Task:
The morning after the hack, your agent books the **cheapest Tatkal ticket from Roorkee to Haridwar** for the next day, for one member of your team.

1. Hands-free. Schedule the run before code freeze. At the right time, your agent logs in to IRCTC, finds the train, books the ticket and saves the proof. The only thing you may do is approve the final payment.
2. Pre-filled details. Keep that passenger's details somewhere your agent can read them: a file, a note, anything.
3. Know your window. The Tatkal window opens at 11:00 sharp.
4. It's on us. If your submission "qualifies", we refund the booking amount.

What You Build:
One generic harness, many skills. The harness sees the screen, decides, acts, checks the result and loops until the task is done or it needs help. A new task should need a new skill, not new code.

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Every harness must have:

1. **Hands:** shell commands, file read and write, browser control, and mouse and keyboard on any desktop app.
2. **Eyes:** screenshots, plus any structure it can get, like the page DOM or terminal output.
3. **The Loop:** plan, act, observe, verify, repeat.
4. **Skills:** a folder of skill files that teach it specific jobs.
5. **Triggers:** start a run from a chat command, a schedule (”at 10:59:30”) or an event (”when an email arrives”).
6. **Human Handoff:** when it is stuck, facing a judgement call or approving a payment, it pauses, asks a person through its own channel, and resumes.
7. **Run Log:** every step saved with a screenshot and the reason for the action, so judges can replay the run.

Bonus Credit for:

1. **Self-improving skills:** after a run, the agent writes what it learned back into the skill.
2. **Skills from a demo:** shown or told a task once, the agent drafts its own skill.

Note: You get only two windows to book, so expect things to go wrong. IRCTC slows down and throws errors right after 11:00, bot checks can block automated browsers (use your own hotspot, not shared Wi-Fi), and your model API can hit its rate limit. Rehearse the whole flow beforehand, and teach your agent how to retry and recover at every step.

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What To Submit:

The hack ends at 09:00 and Tatkal opens at 11:00, so your code and everything else freezes at 09:00. After that, you can edit only one Google Drive folder.

1. **Submission form:** the organizing team will share it.
2. **Demo video:** walk us through your architecture and show what amazing your agent can do, all upon your creativity.
3. **Google Drive folder**, containing:

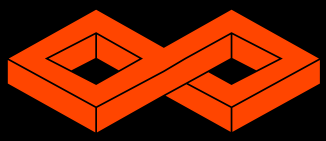
- the run log.
- a screenshot of the ticket with its PNR.
- a screen recording of the whole booking, start to finish, plus a phone video of the untouched device for the same time.
- the IRCTC skills your agent wrote.

How We Judge:

1. **Qualify:** a correct Tatkal ticket, booked hands-free, with proof in your Drive folder, satisfying below TnC.
2. **Final round:** qualifying teams get live tasks across their device, to show how general the agent really is.
3. **Tie-break:** fewer human handoffs, then better architectural judgement and user experience.

Terms and Conditions:

1. The ticket must be the cheapest Tatkal ticket from Roorkee to Haridwar, with the passenger's details correct.



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2. Hands-free means no human intervention except approving the final payment. Any other intervention costs points.
3. Your agent must be generic: not browser-only, and not built or trained just for IRCTC. In the final round you'll get tasks across your whole device.
4. Use any model and any API.
5. One booking per team, from a team member's own IRCTC account. No multiple or borrowed accounts.
6. No code changes after 09:00. Only your Drive folder may change.
7. Your agent runs on your own device and accounts, so you are responsible for what it does.